Understanding Exploratory Data Analysis

What is Exploratory Data Analysis (EDA)?

Exploratory Data Analysis (EDA) is the process of examining a dataset to understand its main characteristics before performing any detailed analysis or building models. It involves summarizing the data using descriptive statistics, identifying missing values and outliers, and creating visualizations such as histograms, box plots, and scatter plots to uncover patterns, trends, and relationships. EDA is open-ended and does not start with a fixed hypothesis. Instead, analysts use it to get familiar with the data, ask questions, and generate insights that guide further investigation.

What is Confirmatory Data Analysis?

Confirmatory Data Analysis (CDA) is a more formal, hypothesis-driven approach to analyzing data. Unlike EDA, which explores data without predetermined assumptions, CDA begins with a specific hypothesis and uses statistical tests to determine whether the data supports or rejects that hypothesis. Techniques such as t-tests, chi-square tests, ANOVA, and regression analysis are commonly used in CDA to measure statistical significance and validate findings with a defined level of confidence.

Main Difference Between EDA and CDA

The main difference between the two is their purpose and approach. EDA is exploratory and flexible, designed to discover what the data contains without any prior assumptions. CDA is confirmatory and structured, designed to test specific claims and validate hypotheses using formal statistical methods.

Example: When EDA Would Be Useful

EDA would be useful when a retail company receives a new dataset containing customer purchase history. Before building a recommendation model, analysts would use EDA to examine the distribution of purchase amounts, identify any missing or inconsistent data, spot outliers, and visualize which product categories are most popular. This exploration helps the team understand the data quality and structure before deciding on the next steps.

Example: When CDA Would Be Useful

CDA would be useful when a pharmaceutical company wants to determine whether a new drug is more effective than a placebo. After conducting a clinical trial, analysts would use a hypothesis test (such as a t-test) to compare the outcomes of the treatment and control groups. The goal is to confirm, with statistical confidence, whether the observed difference in effectiveness is significant or simply due to chance.